MACHINE LEARNING

CORE CONCEPTS HANDBOOK

Model

A **Model** in Machine Learning is a mathematical function or algorithm that learns patterns from data and makes predictions.

Simple Meaning:
A model is a system that takes input (features) and produces output (prediction).

Example

Suppose we want to predict whether a ball is:

Tennis or Cricket

Features:

- Weight
- Surface Type

The model learns from previous examples and creates a decision rule like:

If weight < 80 AND surface = smooth → Tennis
Else → Cricket

This learned rule is the **model**.

Mathematical View

In Linear Regression:

```
y = mx + b
```

The equation itself becomes the model.

In Logistic Regression:

```
P(y=1) = 1 / (1 + e^-(wx + b))
```

The learned weights (w, b) define the model.

Important Point

Model = Learned Pattern
Not just code.
It is the trained mathematical representation.

Training

Training is the process of teaching the model using training data so that it learns patterns.

During training:

- Model sees input data (X)
- Model compares predicted output with actual output (y)
- Model adjusts internal parameters (weights)
- Error is minimized

Example

Training Dataset:

Weight	Surface	Type
60	Smooth	Tennis
90	Rough	Cricket

Model learns:
Smooth + Low Weight → Tennis
Rough + High Weight → Cricket

That learning process is **training**.

During training:

1. Model makes prediction

2. Calculate error (Loss Function)
3. Adjust parameters (Gradient Descent)
4. Repeat many times

Code Example

```
model.fit(X_train, y_train)
```

Here:

- X_train → features
- y_train → labels
- fit() → training process

Important Concept

Training data must contain:

- Inputs (X)
- Correct outputs (y)

Without correct outputs → cannot train supervised model.

Testing

Testing is the process of evaluating how well the trained model performs on unseen data.

The model has never seen this data before.

Testing checks:
Can model generalize to new data?

Example

After training ball classifier, we give new ball:

Weight = 70
Surface = Smooth

Model predicts: Tennis

If correct → good model
If wrong → model needs improvement

Code Example

```
predictions = model.predict(X_test)
accuracy = model.score(X_test, y_test)
```

Here:

- X_test → unseen data
- score() → testing accuracy

Difference Between Training and Testing

Term	Purpose	Data Used
Training	Learn patterns	Training dataset
Testing	Evaluate performance	Unseen dataset

Real-Life Analogy

Imagine preparing for exams.

Training:
You study from textbook and solve practice problems.

Testing:

If you only memorize examples → Overfitting
If you understand concepts → Good generalization

Important Related Terms

Parameters

Values learned during training.

Example:

- Slope in Linear Regression
- Weights in Neural Network

Hyperparameters

Values set before training.

Example:

- K in KNN
- Learning rate
- Max depth in Decision Tree